The Cauchy-Schwartz Equalities

Corbin Smith
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Previously we proved the Cauchy-Schwartz inequality and variant of it. Now we will prove both simultaneously by establishing an equality valid in any vector space equipped with a quadratic form.

Suppose we have a vector space $\mathbb{V}$ over a field $\mathbb{F}$, equipped with a function $Q(\mathbf{x})$ (called a quadratic form for reasons about to be explained) such that $Q(\lambda\mathbf{x}) = \lambda^2 Q(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{V}$. We call the process of applying Q to $\mathbf{x}$ as **measuring** $\mathbf{x}$, and $Q(\mathbf{x})$ is the **quadrance** of $\mathbf{x}$. These conditions mean that when we have a basis then Q is a homogeneous quadratic expression in the coefficients of its argument, and therefore can be expressed as $Q(\mathbf{x}) = \mathbf{x}^T M\mathbf{x}$ for some unique symmetric matrix M. We refer to M as the **metric tensor** of our quadratic form. This insight lets us recognize that the **dot product** of $\mathbf{x}$ and $\mathbf{y}$ computed as $\mathbf{x}\cdot\mathbf{y} = (Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}))/2 = \mathbf{x}^T M\mathbf{y}$ is a symmetric bilinear expression in $\mathbf{x}$ and $\mathbf{y}$. We define $\mathbf{x}$ to be **perpendicular** to $\mathbf{y}$ exactly when $\mathbf{x}\cdot\mathbf{y} = 0$ because this makes the Pythagorean true with any dot product we pick. For any Q, we can determine the coefficients of M by making $\binom{n+1}{2}$ measurements of all the basis vectors and their pairwise sums.

Exercise 1 Show that if we have a basis for our vector space and M^{-1} exists then we can determine the coefficients of any vector simply by knowing how it dots with each of our basis vectors, without constructing some sort of physical coordinate system.

Frequently it is helpful or required to extend the comparisons of our metric tensor to $\bigwedge\mathbb{V}$. In order to do this we will construct an algebra $GA(\mathbb{V}, Q(\mathbf{x}))$ which contains $\bigwedge\mathbb{V}$ but is also sensitive to $Q(\mathbf{x})$. This algebra is called the **Geometric Algebra** of the pair $(\mathbb{V}, Q(\mathbf{x}))$. We could do it using quotients $GA(\mathbb{V}, Q(\mathbf{x})) = \bigotimes\mathbb{V}/(\mathbf{x}\otimes\mathbf{x} - Q(\mathbf{x}))$ similar to constructions of $\bigwedge\mathbb{V} = \bigotimes\mathbb{V}/(\mathbf{x}\otimes\mathbf{x})$, but an axiomatic approach would be much simpler.

Before I state these axioms, let me just say that they are an attempt to generalize the idea of scalar multiplication and division to higher dimensional objects, at the expense of commutativity. Therefore we will denote multiplication of elements of this algebra by simply concatenating them. We will also insist that the characteristic of our field is not 2. Furthermore we will insist for $Q(x) = \mathbf{x}^T M\mathbf{x}$ that M^{-1} exists.

Our axioms for $GA(\mathbb{V}, Q(\mathbf{x}))$ are

A1: We can add together different types of objects We have that $GA(\mathbb{V}, Q(\mathbf{x}))$, written in shorthand as GA, is a ring with unity containing subsets $\mathbb{F} = GA_0$ and $\mathbb{V} = GA_1$.

A2: Addition and multiplication are compatible with scalar arithmetic The ring identities of $GA(\mathbb{V}, Q(\mathbf{x}))$ are the the identities of $\mathbb{F}$; $0_\mathbb{F}$ and $1_\mathbb{F}$. We also have that $\mathbb{F}$ is contained in the center of GA; scalars commute with everything.

A3: The algebra is generated by vectors Every element in our algebra can be expressed as a linear combination of a product of vectors $GA = \langle\mathbb{V}\rangle$, with the convention that an empty product is the multiplicative identity.

A4: Every vector squares to a scalar For any $\mathbf{x} \in \mathbb{V}$, we have that $\mathbf{x}^2 = Q(\mathbf{x})$.

From these axioms and the properties of scalar multiplication of vectors we can conclude that $0_{\mathbb{F}} = \mathbf{0}_{\mathbb{V}}$ our scalar 0 is the same as our zero vector. Notice that if $Q(\mathbf{x}) = 0$ for all $\mathbf{x} \in \mathbb{V}$ then $GA = \bigwedge \mathbb{V}$. There is still a sense in which this is true even when $Q(\mathbf{x}) \neq 0$ for some $\mathbf{x} \in \mathbb{V}$, but in order to elucidate that we need to do a little work. One thing that might be surprising is that this algebra provides us with a geometrically meaningful way define vector division.

Exercise 2 Observe the *fundamental identity*:

$$\mathbf{xy} = \frac{1}{2}(\mathbf{xy} + \mathbf{yx}) + \frac{1}{2}(\mathbf{xy} - \mathbf{yx}).$$

Prove the following form of the *polarization identity*:

$$\mathbf{x} \cdot \mathbf{y} = \frac{1}{2}(\mathbf{xy} + \mathbf{yx}),$$

use it to demonstrate the critically important *palindrome identity*:

$$\mathbf{xyz} - \mathbf{zyx} = \mathbf{yzx} - \mathbf{xzy} = \mathbf{zxy} - \mathbf{yxz},$$

as well as the equally important *swap lemma*:

$$\mathbf{xy} = 2\mathbf{x} \cdot \mathbf{y} - \mathbf{yx}.$$

Conclude that two vectors anticommute precisely when they are perpendicular.

Now we can recover the wedge product and exterior powers of our vector space. Given a set of k vectors $\mathbf{x}_1, ..., \mathbf{x}_k$, we define the **wedge product** of the set to be

$$\mathbf{x}_1 \wedge ... \wedge \mathbf{x}_k = \frac{1}{k!} \sum_{\sigma \in \mathfrak{S}_k} sign(\sigma) \mathbf{x}_{\sigma(1)} ... \mathbf{x}_{\sigma(k)}.$$

Expressions which can be written in this form are called **k-blades**. The difficulties of this definition that occur when the characteristic is nonzero can be neutralized, but wont be here. Blades that are not scalars or vectors are typically denoted with upper case bold letters. The most important case to pay attention to is $\mathbf{x} \wedge \mathbf{y} = \frac{1}{2}(\mathbf{xy} - \mathbf{yx})$, which means the fundamental identity can be restated as $\mathbf{xy} = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \wedge \mathbf{y}$.

Exercise 3 Show that our definition of wedge product is distributive in each argument, and that the wedge product of a set is zero if a vector appears twice. In order to truly conclude that this is wedge as we know it, we will defer to the next exercise for a proof of associativity. Prove that a set of vectors is linearly dependent exactly when its wedge product is zero and that two blades factor into sets of vectors which span the same subspace exactly when they are scalar multiples of each other.

The previous exercise lets us interpret blades as subspace representatives if we can show that we can wedge blades representing disjoint subspaces (except for the zero vector of course) together in order obtain a blade representing the span of the two subspaces, and that this operation is associative. By induction, it would be enough to prove that $\mathbf{x}\wedge(\mathbf{y}\wedge\mathbf{z}) = (\mathbf{x}\wedge\mathbf{y})\wedge\mathbf{z}$.

Exercise 4 Using the signed sum over permutations definition of wedge and the palindrome identity, show that the three following equalities

$$\mathbf{x}\wedge\mathbf{y}\wedge\mathbf{z} = \frac{1}{2}(\mathbf{xyz} - \mathbf{zyx}),$$

$$\mathbf{x}\wedge\mathbf{y}\wedge\mathbf{z} = \frac{1}{4}(\mathbf{xyz} - \mathbf{xzy} + \mathbf{yzx} - \mathbf{zyx}) = \frac{1}{2}(\mathbf{x}(\mathbf{y}\wedge\mathbf{z}) + (\mathbf{y}\wedge\mathbf{z})\mathbf{x}),$$

$$\mathbf{x}\wedge\mathbf{y}\wedge\mathbf{z} = \frac{1}{4}(\mathbf{xyz} - \mathbf{y}\,\mathbf{x}\,\mathbf{z} + \mathbf{z}\,\mathbf{x}\,\mathbf{y} - \mathbf{zyx}) = \frac{1}{2}((\mathbf{x}\wedge\mathbf{y})\mathbf{z} + \mathbf{z}(\mathbf{x}\wedge\mathbf{y})).$$

Let $\mathbf{B}$ be a k-blade, define $\mathbf{a}\wedge\mathbf{B} = \frac{1}{2}(\mathbf{aB} + (-1)^k\mathbf{Ba})$ (actually this is a theorem) as well as $\mathbf{B}\wedge a = \frac{1}{2}(\mathbf{Ba} + (-1)^k\mathbf{aB})$, and conclude that wedge is associative. For a proof that does not involve an induction, prove the identity:

$$\mathbf{acB} - \mathbf{Bca} = \mathbf{Bac} - \mathbf{caB}$$

and apply our most previous definitions.

At this point there is still so much left to do, but at the same so many benefits to collect. By writing the fundamental identity backwards we can transform it from $\mathbf{xy} = \mathbf{x}\cdot\mathbf{y} + \mathbf{x}\wedge\mathbf{y}$ to $\mathbf{yx} = \mathbf{x}\cdot\mathbf{y} - \mathbf{x}\wedge\mathbf{y}$. Multiplying these two expressions together and simplifying both sides gives us the **Cauchy-Schwartz Equality**

$$\mathbf{x}^2\mathbf{y}^2 = (\mathbf{x}\cdot\mathbf{y})^2 - (\mathbf{x}\wedge\mathbf{y})^2$$

which is valid in every dimension, with any* quadratic form, over any* field (excluding the cases we have already discussed).

Define **blue space** to be the GA of the space spanned by $\mathbf{b}_1$ and $\mathbf{b}_2$ with $\mathbf{b}_1^2 = \mathbf{b}_2^2 = 1$ and $\mathbf{b}_1\cdot\mathbf{b}_2 = 0$. Notice that $(\mathbf{b}_1\mathbf{b}_2)^2 = -1$. Then for $\mathbf{x} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2$, $\mathbf{y} = y_1\mathbf{b}_1 + y_2\mathbf{b}_2$ we have that $\mathbf{xy} = x_1y_1 + x_2y_2 + (x_1y_2 - x_2y_1)\mathbf{b}_1\mathbf{b}_2$ and consequently

$$(x_1^2 + x_2^2)(y_1^2 + y_2^2) = (x_1y_1 + x_2y_2)^2 + (x_1y_2 - x_2y_1)^2.$$

Define **red space** to be the GA of the space spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$ with $\mathbf{r}_1^2 = -\mathbf{r}_2^2 = 1$ and $\mathbf{r}_1\cdot\mathbf{r}_2 = 0$. Notice that $(\mathbf{r}_1\mathbf{r}_2)^2 = 1$. Then for $\mathbf{x} = x_1\mathbf{r}_1 + x_2\mathbf{r}_2$, $\mathbf{y} = y_1\mathbf{r}_1 + y_2\mathbf{r}_2$ we have that $\mathbf{xy} = x_1y_1 - x_2y_2 + (x_1y_2 - x_2y_1)\mathbf{r}_1\mathbf{r}_2$ and consequently

$$(x_1^2 - x_2^2)(y_1^2 - y_2^2) = (x_1y_1 - x_2y_2)^2 - (x_1y_2 - x_2y_1)^2.$$

The reader is invited to write down this equality in three dimensional coordinates with quadratic forms $B_3(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2$ and $R_3(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2$.